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Q1.(Creating Lists)

```
integer_list = [10, 20, 30, 40, 50]

empty_list1 = []

empty_list2 = list()

mixed_list = [100, "Python", 99.5, True]

print("a) List of 5 integers:")

print(integer_list)

print("\nb) Empty list using []:")

print(empty_list1)

print("\nEmpty list using list():")

print(empty_list2)

print("\nc) List with mixed data types:")

print(mixed_list)

print("\nd) List of 7 zeros using repetition (*):")

print(zero_list)
```

Q2.(Indexing and Negative Indexing)

```
fruits = ["apple", "banana", "mango", "orange", "grape"]

print("First item:", fruits[0])

print("Third item:", fruits[2])

print("Last item:", fruits[-1])
```

```
print("Second last item:", fruits[-2])

index = int(input("Enter the index number: "))

if index >= 0 and index < len(fruits):

    print("Item at index", index, "is:", fruits[index])

else:

    print("Invalid index!")
```

Q3.(Slicing Lists)

```
numbers = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]

print("a) First 4 elements:", numbers[0:4])

print("b) Last 3 elements
```

Q4.(Modifying Lists)

```
colors = ["red", "blue", "green", "yellow"]

print("Original list:", colors)

colors[1] = "purple"

print("After changing 'blue' to 'purple':", colors)

colors[-1] = "black"

print("After changing the last item to 'black':", colors)
```

Q5.(append() and insert())

```
cities = []

city1 = input("Enter the first city: ")

city2 = input("Enter the second city: ")
```

```
cities.append(city1)
cities.append(city2)
cities.append("Mumbai")
cities.append("Pune")
cities.append("Nagpur")
new_city = input("Enter a city to insert at position 2: ")
cities.insert(2, new_city)
print("\nFinal List of Cities:")
print(cities)
```

Q6.(remove(), pop(), and clear())

```
items = [10, 20, 30, 20, 40, 50]
print("Original List:", items)
items.remove(20)
print("\nAfter remove(20):", items)
removed_item = items.pop(3)
print("\nRemoved item using pop(3):", removed_item)
print("List after pop(3):", items)
last_item = items.pop()
print("\nRemoved last item:", last_item)
print("List after removing last item:", items)
items.clear()
print("\nList after clear():", items)
```

Q7.(index() and count())

(index() and count())

Create a list: scores = [85, 92, 78, 92, 65, 92, 88]

Find and print the index of the first occurrence of 92.

Count and print how many times 92 appears.

Take a number as input from the user and check if it exists in the list. If yes,

print its index and count.

Q8.(sort() and reverse())

```
marks = [45, 78, 65, 90, 34, 82, 71]

print("Original List:", marks)

marks.sort()

print("\nAscending Order:", marks)

marks.sort(reverse
```

Q9.(extend() vs append())

```
list1 = [1, 2, 3]

list2 = [4, 5, 6]

list_extend = list1.copy()

list_extend.extend(list2)
```

```
print("Using extend():")
print(list_extend)
list_append = list1.copy()
list_append.append(list2)
print("\nUsing append():")
print(list_append)
```

Q10.(Mini Project - Combined Concepts)

```
marks = []
for i in range(5):
    mark = float(input(f"Enter marks for Subject {i + 1}: "))
    marks.append(mark)
print("\nMarks List:", marks)
new_mark = float(input("Enter marks for one more subject: "))
marks.append(new_mark)
print("\nUpdated Marks List:", marks)
print("Highest Marks:", max(marks))
print("Lowest Marks:", min(marks))
marks.sort(reverse=True)
print("\nMarks in Descending Order:", marks)
average = sum(marks) / len(marks)
print("Average Marks:", average)
print("Total Number of Subjects:", len(marks))
```